

Assignment 4 – Image Classification

Basic CNN Model

Basic CNN Architecture and Performance

Architecture Overview:

- **Layer 1:** Features a configuration of 16 filters, each 3x3 in size, with stride 1 and padding maintained at 1. Utilizes ReLU for activation followed by batch normalization and pooling operations to reduce dimensionality while retaining key features.
- **Layer 2:** Increases the filter count to 32, preserving the 3x3 size with consistent stride and padding. Continues with ReLU activation, complemented by batch normalization and pooling.
- **Layer 3:** Expands to 64 filters of the same size, employing the established pattern of ReLU activation along with subsequent normalization and pooling steps.

Performance Metrics:

- **Train Loss:** Recorded at approximately 0.496, indicating initial model convergence.
- **Validation Loss:** Measures around 1.270, suggesting room for improvement in model generalization.
- **Test Accuracy:** Achieves about 64.10%, showcasing a solid base model performance.

Advanced Model: All Convolutional Network

Architecture Details:

- **Layers 1 to 3:** Utilizes 96 filters each, with the third layer applying a stride of 2 to effectively down sample the spatial dimensions.
- **Layers 4 to 6:** Filter count increases to 192, with the sixth layer further employing a stride of 2 for additional reduction in spatial dimensions.
- **Layers 7 and 8:** Continue refining feature maps, with the eighth layer using a 1x1 convolution to compact the depth dimensions.
- **Final Layer:** Employs filters equal to the class count (10), culminating in global average pooling to derive class scores.

Performance Overview:

- **Training Loss:** Approximates 0.660, indicating the model's learning progression.
- **Validation Loss:** Approximated at 1.040, reflects the model's adaptability to unseen data.
- **Test Accuracy:** Estimated at 65.63%, a slight improvement over the basic model.

Enhanced Model: All Convolutional Network with Regularization

Architectural Configuration:

- Identical to the standard All Convolutional Network but includes strategic implementation of dropout layers post layers two and five to mitigate overfitting.
- Incorporates data augmentation in the training phase with random horizontal flips, rotations up to 10 degrees, and random resized crops to enhance the robustness of the model.

Comparative Performance:

- **Without Regularization:**
 - **Train Loss:** ~0.662
 - **Validation Loss:** ~1.035
 - **Test Accuracy:** ~65.63%
- **With Regularization:**
 - **Train Loss:** Reduced to ~0.614

- **Validation Loss:** Lowered to ~0.948
- **Test Accuracy:** Improved significantly to ~69.04%

Transfer Learning Integration

Target Model: All Convolutional Network with Regularization

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Transfer Learning Integration

Target Model: All Convolutional Network with Regularization

Performance Metrics:

- **Training Loss:** Approximately 0.760, indicative of challenges in model training.
- **Validation Loss:** Noted as ~1.080, signaling potential overfitting.
- **Test Accuracy:** Approximated at 63.88%, slightly lower due to possible issues in effective transfer learning adaptation.

Conclusion: The All-Convolutional Net with Regularization stands out as the best model, achieving the highest accuracy of 69.04% in testing scenarios.